

Saving for a Rainy Day: Experimental Evidence on Prize Linked Saving and Financial Shocks

Kaden Grace

Dept. of Economics, University of Tennessee - Knoxville

Dissertation Proposal

Household Liquidity Constraints

A Trip to the ER

- Nearly two-fifths of U.S. adults report inability to access \$2,000 within one month
FED Survey, (2025)
- Households may make suboptimal financial decisions: asset liquidation, high-interest borrowing, reducing consumption

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Credit markets costly or inaccessible:

- Limited access, adverse credit ratings, predatory lending
- Example: Tennessee Flex Loans charge 279.5% interest

Why don't households save more?

Determinants of insufficient savings:

- High discount rates (present bias)
- Low risk aversion
- External income constraints
- Serially correlated shocks (bad luck)
- Financial illiteracy

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Behavioral Intervention: Prize-Linked Savings Accounts

Design:

- Deposits function as lottery entries with stochastic returns
- Principal protected – “no-lose lottery”
- Win probability proportional to deposit

Target population:

- Individuals with high discount rates, low risk aversion
- Legal in 33 U.S. states following 2014 American Savings Promotion Act

Ambiguous Ex-Ante Effects on Total Saving

Benefits:

- Increased savings participation among under-savers
- Enhanced behavioral engagement
- **Appealing to the most vulnerable households** Gertler et al, (*NBER WP*, 2023)

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- Crowding out traditional savings
- Foregone compound interest
- **Portfolio substitution without net savings increase** Jindapon et al., (*GEB*, 2022); Filiz-Ozbay et al. (*J. Pub. Econ*, 2015)

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Research Question: Does access to a PLSA increase total precautionary savings for households facing financial shocks?

Preview of Results

PLSAs cause portfolio substitution without increasing total savings.

- ① Crowding out** traditional savings by 33.6–38.5%
 - Risk-seeking participants: 93.7% crowd out
- ② No net savings increase** for any risk preference group
- ③ Earnings decline** 9.1–17.8% from foregone compound interest

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This paper's contribution:

- Heterogeneous treatment effects by risk preference
- First test of PLSA crowding out for *very high-yield* savings (16%)

Round

Remaining time: 29

Instructions: Your task is to allocate your tokens.
You can place your tokens in the **Lottery Account** and **convert tokens to dollars**.

Any tokens you do not use will be saved in your **Bank Account** until the next round and earn 16% interest.

Remember, the only way to earn real money in this experiment is by converting your tokens into dollars.

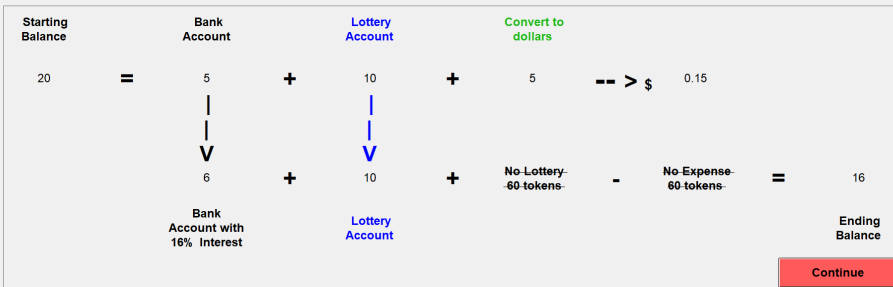
Starting Balance		Bank Account		Lottery Account		Convert to dollars		
20	=	0	+		+		-- >	\$ 0.00
		↓ V ↓		↓ V ↓				
		0	+	0	+	60 tokens Lottery Chance: %	-	60 tokens Expense Chance: 10%
		Bank Account with 16% Interest		Lottery Account		0.00		Ending Balance

Calculate
Submit

Round

1

Results



Round	1	2	3	4	5	6	7	8	9	10	Total
Earnings	\$0.15	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.15

Experimental Design

- Dynamic life-cycle framework: 10 periods, each participant plays two treatments
- Exog. income 20 tokens per period
- Portfolio allocation: consumption, bank savings (16%), or PLSA **CARA Utility**
- Shock: 10% probability of 60-token expense per period
- PLSA expected return equals bank account (16%)

All Parameters**Why an experiment?**

Experimental Design

Three within-subject treatments:

- Control: No PLSA access
- High-prize PLSA: 120-token prize
- Low-prize PLSA: 60-token prize

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Session Type	N
No Prize/High Prize	56
High Prize/Low Prize	49
No Prize/Low Prize	52
Total	157

	Mean	St. Dev.	Min.	Max
Assets	88.34	95.59	-294	680
Standard Savings	57.62	85.19	-294	624
PLSA Savings	20.82	58.43	0	680
Tokens Consumed	9.92	26.81	0	455
Optimal Consumption	18.98	38.82	0	311
Dev. from Optimal Cons.	-9.06	42.68	-311	232
Abs. MPC Error	0.21	0.25	0	1
Expense Incurred	0.09	0.29	0	1
High Prize PLSA	0.33	0.47	0	1
Low Prize PLSA	0.32	0.47	0	1
Any PLSA	0.66	0.48	0	1

Estimating Equation

$$\text{Outcome}_{ist} = \alpha + \underbrace{\beta_1 \cdot \text{PLSA High}_{is}}_{\substack{=1 \\ \text{if PLSA} \\ \text{offered} \\ \text{(High Prize)}}} + \underbrace{\beta_2 \cdot \text{PLSA Low}_{is}}_{\substack{=1 \\ \text{if PLSA} \\ \text{offered} \\ \text{(Low Prize)}}} + \underbrace{\delta \cdot \text{Second}_{is}}_{\substack{=1 \\ \text{if played} \\ \text{second} \\ \text{in session}}} + \bar{\gamma} \cdot \underbrace{\bar{X}_i}_{\substack{\text{part.} \\ \text{demog.}}} + \underbrace{u_{ist}}_{\substack{\text{error} \\ \text{corr. w/in} \\ \text{participants}}}$$

$$\text{where Outcome}_{ist} \in \left\{ \begin{array}{l} \text{Bank Savings}_{ist} \\ \text{Bank + PLSA Savings}_{ist} \\ \text{Earnings}_{is} \\ |\text{MPC Error}_{ist}| \end{array} \right\}$$

Crowding Out Bank Savings

- Bank account savings decline 33.6–38.5%
- Risk-seeking:** 93.7% crowding out
- Risk-averse:** 30.4–40.3% crowding out

Combined Risk Cat.

	All	Seeking	Neutral	Averse	V. Averse
High Prize PLSA	-27.17*** (7.25)	-27.47 (16.65)	-9.11 (14.91)	-31.89*** (9.32)	-21.34 (65.90)
Low Prize PLSA	-23.68*** (7.41)	-48.61** (19.21)	-1.88 (17.63)	-24.05** (9.17)	-36.85 (56.31)
Second	7.53 (5.07)	14.39 (11.82)	5.39 (9.41)	7.10 (6.93)	22.70 (33.41)
Constant	70.56*** (5.92)	51.86*** (10.68)	45.36*** (13.57)	79.17*** (7.38)	93.52 (54.55)
N	3140	480	480	2020	160

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No Increase in Total Savings

- Total savings unchanged
- PLSAs induce **portfolio reallocation** without increasing savings

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High Prize PLSA	5.72 (7.03)	20.30 (16.56)	4.85 (15.24)	3.13 (8.94)	-14.72 (61.48)
Low Prize PLSA	6.90 (8.25)	-19.08 (20.62)	43.45* (22.57)	6.09 (10.14)	-27.79 (55.24)
Second	8.49 (5.57)	1.23 (11.49)	9.94 (13.40)	12.19 (7.48)	17.78 (32.79)
Constant	70.06*** (6.18)	58.83*** (14.26)	42.58*** (14.63)	76.81*** (7.50)	98.44 (53.56)
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Lower Earnings from Foregone Interest

- Realized earnings decline
9.1–17.8%
- Opportunity cost: foregone
compound interest
- Utility from lottery experience
must compensate

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	All	Seeking	Neutral	Averse	V. Averse
High Prize PLSA	-0.397* (0.215)	-0.333 (0.634)	-0.874 (0.552)	-0.305 (0.263)	0.744 (0.490)
Low Prize PLSA	-0.774*** (0.233)	-1.080 (0.647)	-0.506 (0.786)	-0.793*** (0.286)	0.782 (0.891)
Second	-0.353* (0.184)	-0.374 (0.568)	-0.350 (0.564)	-0.322 (0.214)	0.465 (0.808)
Constant	4.350*** (0.178)	4.678*** (0.410)	4.523*** (0.568)	4.287*** (0.214)	2.227** (0.801)
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Enhanced Consumption Smoothing

- Consumption 13.1–19.2% closer to optimal
- **Risk-seeking:** 52.2–60.6% improvement
- **Risk-averse:** Null effect

Combined Risk Cat.

	All	Seeking	Neutral	Averse	V. Averse
High Prize PLSA	-0.035* (0.018)	-0.094* (0.051)	-0.129** (0.050)	-0.001 (0.019)	0.030 (0.057)
Low Prize PLSA	-0.055*** (0.018)	-0.086 (0.056)	-0.142*** (0.043)	-0.029 (0.020)	0.010 (0.025)
Second	-0.026** (0.013)	-0.025 (0.039)	-0.039 (0.033)	-0.026* (0.014)	-0.045* (0.019)
Constant	0.252*** (0.017)	0.329*** (0.033)	0.332*** (0.050)	0.219*** (0.020)	0.172*** (0.019)
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Summary of findings:

- PLSAs crowd out bank savings by 33.6–38.5%, total savings are unchanged
- **Stronger effect on risk-seeking individuals**
- Earnings decline 9.1–17.8% from foregone returns

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Policy implications:

- **For Banks:** PLSAs can increase profit; crowd-out is sensitive to size of prize
- **For Gov't:** PLSAs appeal to the most vulnerable households
- **Overall: PLSAs are most profitable and engaging when marketed towards risk-seeking, unbanked households.**

Thank You!

kgrace7@vols.utk.edu

Why use a lab experiment?

- Observational data limitations:
 - Identification of exogenous shocks problematic
 - Complete household portfolio allocation unobserved
- Experimental advantages:
 - Random treatment assignment ensures exogeneity
 - Precise parameterization of choice environment
 - Full observability of portfolio decisions

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Tokens	Dollars
10	\$ 0.29
20	\$ 0.54
30	\$ 0.78
40	\$ 0.99
50	\$ 1.18
60	\$ 1.35
70	\$ 1.51
80	\$ 1.65
90	\$ 1.78
100	\$ 1.90

Tokens	Dollars
110	\$ 2.00
120	\$ 2.10
130	\$ 2.18
140	\$ 2.26
150	\$ 2.33
160	\$ 2.39
170	\$ 2.45
180	\$ 2.50
190	\$ 2.55
200	\$ 2.59

Tokens	Dollars
210	\$ 2.63
220	\$ 2.67
230	\$ 2.70
240	\$ 2.73
250	\$ 2.75
260	\$ 2.78
270	\$ 2.80
280	\$ 2.82
290	\$ 2.83
300	\$ 2.85

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Parameter	Value
Income (y_t)	20 tokens
Discount Rate (β)	1
Interest rate (r)	16%
PLS Prize (Q)	None, 60, or 120 tokens
Expense (K)	60 tokens
Probability(Expense) (p)	10%
CARA Risk Aversion Parameter	0.01
Maximum Consumption per Period	\$3

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Earnings, Seeking/Neutral

Combined [Back](#)

	Full Sample	Not Risk Averse	Averse	V. Averse
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Bank Account, Seeking/Neutral Combined

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Total Savings, Seeking/Neutral Combined

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